

***elfs*: An R Package to Promote the Intentional Selection of Scoring Method for Psychological Scales**

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Abstract

Psychologists typically rely on sum or mean scores to approximate latent factors such as self-esteem or depression. They often do so (a) without providing relevant information about validity (such as fit statistics or evidence of measurement invariance) and (b) without evaluating the appropriateness of sum scores versus latent factor scores for a given scale. We discuss the relevant merits and tradeoffs of each scoring approach, then introduce the *elfs* package (*Evaluating and Estimating Latent Factor Scores*) for R. *Elfs* requires simpler inputs than existing packages for latent factor estimation and produces clearer outputs in the form of a step-by-step flowchart with color-coded tables that researchers can use to evaluate which scoring method best suits their data. By removing barriers to intentional measurement practices, we hope to improve researchers' ability to accurately tap constructs of interest as well as readers' ability to clearly evaluate measurement choices.

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keywords: measurement; sum scores; latent factor scores; measurement invariance; structural equation modeling

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Psychological constructs such as extraversion, depression, and self-esteem are unobservable and cannot be measured directly with physical instruments. Researchers approximate these latent constructs by measuring indicators (e.g., survey responses, observed behaviors) and combining these indicators into a score that represents the construct of interest. Early psychological research overwhelmingly relied on sum scoring (or mean scoring; Embretson, 2004; Wilson et al., 2006), given its mathematical soundness under many conditions (as established by Classical Test Theory; CTT; Lord et al., 1968) and its practical ease (you could sum score responses on a napkin).

Despite these advantages, advances in latent construct measurement suggest that latent factor scores from models like Item Response Theory (IRT) or Structural Equation Modeling (SEM) have various desirable properties not afforded by sum scoring (e.g., Brown & Maydeu-Olivares, 2013; Reise et al., 2005; Reise & Waller, 2009; Thissen et al., 1995).¹ Because of these properties, psychometricians often use these approaches (Edelen & Reeve, 2007; Reise & Henson, 2003; von Davier, 2010; but, see Sijtsma et al., 2024); however, empirical psychological researchers still overwhelmingly rely on sum scoring to represent latent variables (Blanchin et al., 2011; Foster et al., 2017; Gortler et al., 2016; Sijtsma, 2012, p. 5).

These practices are largely pragmatic, rather than theoretically motivated (Flake & Fried, 2020; McNeish, 2024)—the “barrier to entry” is lower for sum scoring than for IRT or SEM,

¹ The scope of this paper is broad, so we combine two concepts under the broader term “latent factor scoring” for simplicity of exposition. This term includes one-step approaches that embed the latent variable directly into a model and two-step approaches that predict latent variable values for each individual and then pass those manifest scores on to another subsequent analysis like a regression. There are technical distinctions between these approaches, but for our purposes, they both represent model-based scoring using a parametric measurement model.

both for understanding the mathematical underpinnings of these methods and for writing code to extract latent factor scores using these methods. Summing or averaging responses is not the problem per se because this approach can be justified following psychometric analyses (e.g., Stochl et al., 2022). However, recent criticisms have noted that sum or mean scores are often used uncritically or perfunctorily without supporting justification or analysis to demonstrate that properties of items or the scale warrant such an approach (e.g., dimensionality assessment, invariance testing).

For instance, several review studies across health, social, and clinical psychology consistently report that less than 30% of studies provide validity evidence for their scores and scoring decisions (Crutzen & Peters, 2017; Flake et al., 2017; Higgins et al., 2024; Maassen et al., 2025; Shaw et al., 2020; Weidman et al., 2017). Therefore, the issue appears to run deeper than philosophically abstract debates among methodologists about the relative statistical nuances of sum scores or factor scores. Instead, the deeper issue is that few empirical studies support their decisions with empirical evidence identifying the best scoring approach for their data, which can harm the validity, precision, and reproducibility of inferences in psychological research.

The goal of this paper is to lower barriers for psychologists to adopt modern psychometric practices and identify the best possible scoring approach. We summarize the merits of both latent factor scores and sum scoring and provide a practical flowchart for determining which method better suits various measurement situations. Then, we describe existing challenges for (a) evaluating the appropriateness of latent factor scoring versus sum scoring and (b) estimating and interpreting latent factors within an SEM framework. Finally, we introduce the new R package *elfs* (short for *Evaluation and Estimation of Latent Factor Scores*), which eases these challenges.

Weighing the Merits of Latent Variable Estimation and Sum Scores

Both latent factor scores and sum scores have distinct advantages. We consider each, then reflect on the factors that should (and should not) influence researchers' choices between these scoring methods.

Advantages of Sum Scores

A primary reason for sum scoring is that the scoring procedure for combining items into a single score remains constant and is not subject to sample-to-sample variability. This can allow scores produced from the same scale in different studies to be directly comparable. For instance, if a scale is composed of a checklist of mental health symptoms and the scale is sum scored, it is straightforward and intuitive to compare the scores of two people with a score of 8 from different studies (i.e., they both endorsed that eight symptoms are present).

Relatedly, sum scores can be advantageous if data contain limited or noisy information like a small sample or low reliability (Uanhoro, 2019; Strauss, 2027). In such situations, a sum score may add rigidity and structure to the scoring process and reduce reliance on sample-specific estimates that may be prone to high sampling variability or non-negligible bias (Widaman & Revelle, 2023). The thought process in this case is that sum scores may not necessarily be ideal, but the restrictions they impose are preferable to alternative methods that rely on quantities estimated from the data that may have a low signal-to-noise ratio (Rhemtulla & Savalei, 2025).

Another potential benefit is that sum scores under CTT do not necessarily assume a particular underlying factor structure or even that latent variables exist at all (Borsboom, 2005). This means that the distinction between reflective measurement (i.e., the latent variable causes item responses) and formative measurement (i.e., the latent variable is caused by item responses)

is less relevant (Rhemtulla et al., 2020). This property can be helpful for situations in which a composite score is merely an index that seeks to combine multiple pieces of information rather than a score representing a particular construct (Sijtsma et al., 2024). CTT also permits estimating reliability of a composite or index, regardless of whether the score is intended to represent a particular psychological construct.

Advantages of Latent Variable Estimation

Latent factor scores weigh each item differently so that the items most strongly related to the theorized construct are weighed more heavily than items more weakly related to the construct. For instance, if two people both endorse that eight symptoms are present, they may not necessarily have the same value on the underlying construct of mental health if Person A endorsed the eight mildest symptoms and Person B endorsed the eight most severe symptoms. A latent factor score would potentially differentiate these two response patterns and provide different scores for each person whereas a sum score generally would not.

In simulated data where a person's "true" construct score is known, latent variable scores tend to correlate more strongly with the true construct score than sum scores (Beauducel & Leue, 2013; Ferrando & Lorenzo-Seva, 2021). This property potentially leads to gains in performance if scores are intended to (a) classify individuals (e.g., whether someone has a mental health condition, evaluating a job applicant has sufficient expertise; McNeish, 2023; Morgan-López et al., 2022) , (b) evaluate group mean differences (Ghisetta & Mason, 2025; Gilbert et al., 2026; Schmalbach et al., 2025; Steinmetz, 2013) or (c) understand growth trajectories (Edwards & Soland, 2025; Gorter et al., 2020; Kuhfeld & Soland, 2022; Soland et al., 2025).

Latent factor scores can also maintain advantages in contexts where scales are multidimensional such that more than one construct is being measured (Pruzek & Frederick,

1978, p. 262). In this context, latent factor scores are able to incorporate factor correlations into the scoring process whereas sum scores generally cannot (Butler, 1983). An extension of this idea applies to models with residual correlations, which can be viewed as another manner in which items may measure more than one construct (Ip, 2010). In such cases, latent scores tend to be able to incorporate the residual correlations whereas sum scores cannot (Ferrando et al., 2025).

Given that latent factor scores are model-based, it is straightforward to evaluate measurement invariance to ensure that scores function similarly across different groups of people or timepoints (e.g., Fried et al., 2016; Thissen, 2025; van de Schoot et al., 2012). If scores should be calculated conditionally such that items may contribute different information about the construct for different groups of people (Curran et al., 2018; Mislevy et al., 1992), this can also be incorporated into the scoring process with approaches like multiple indicator-multiple cause (MIMIC; Bauer, 2023; Jöreskog & Goldberger, 1975) models, moderated nonlinear factor analysis (MNLFA; Bauer, 2017; Curran et al., 2014)², or scoring from models that assume only partial measurement invariance (Lai et al., 2017; Millsap & Kwok, 2004).

Sum scores have limited ability to assess invariance and less flexibility to incorporate potential differences in how items function across groups or time by virtue of their constrained scoring system (Gonzalez & Pelham, 2021; Neale, 2005; Slof-Op 't Landt et al., 2009; Wilson et al., 2006). Using sum scores in the presence of non-invariant items can potentially create large biases (Steinmetz, 2013; Millsap & Kwok, 2004).

² Note that the appropriateness of these approaches can vary by area of application. For instance, it is perfectly defensible to score a depression scale differently for men and women because symptoms may manifest different for people who identify as different sexes. However, it may be less defensible (or possibly illegal) to score a mathematics ability scale differently for students whose racial identification is white versus black.

Choosing a Method

We are not recommending that psychologists default to any particular scoring strategy. Instead, our perspective is that psychologists should make evidence- and theory-based decisions when they choose a scoring method to represent unobservable constructs or ideas. The frequent reliance on sum scores in empirical studies is not necessarily a problem in and of itself; rather, the primary issue is that there is no underlying evidence or theory to justify or support this choice of scoring in many studies (McNeish & Wolf, 2020; Schmalbach et al., 2025). Instead, sum scores are often used for heuristic reasons like being (a) familiar to readers, (b) easy to calculate, or (c) understandable without training in psychometrics (DiStefano et al., 2009; Grice & Harris, 1998).

The distinction among methods essentially comes down to a classic bias-variance tradeoff in statistics where researchers must balance optimal estimates in a single sample against good performance across many different samples (e.g., Belkin et al., 2019). Researchers commonly encounter this issue in regression modeling where the best coefficient estimates in a given sample rarely work as well in other samples (readers may be familiar with regularization, which is designed to provide estimates that strike a better balance; e.g., Tibshirani, 1996). The next two sections explain how the bias-variance tradeoff translates to the context of scoring psychological scales.

Choosing Sum Scores

In the context of scoring, sum scores tend to be a lower-variance, higher-bias method. By virtue of having fixed scoring system, the approach is consistent across different samples and the method is unaffected by idiosyncrasies in individual datasets. This property can be desirable, particularly in domains where clear comparison and communication is a central goal (e.g.,

education and mental health; Sijtsma et al., 2024). However, the tradeoff of this fixed scoring system is that the scores may less accurately represent the construct of interest. In other words, the consistency across samples potentially sacrifices some validity because the scores are less closely related to a person's "true" construct score.

Importantly, the extent of this bias is primarily a function of how similarly each item relates to the underlying construct. If items are each similarly related to the underlying construct, then this bias will tend to be small or even negligible, especially if there is a large number of items (Wilks, 1938). Conversely, if items are unevenly related to the underlying construct (i.e., some items have a stronger association with the construct), then the connection between a sum score and the "true" construct score will tend to be weaker because less informative items are treated as interchangeable with more informative items.

Therefore, when items contribute similar amounts of information about the construct, sum scoring is often preferable because the scoring system is consistent and the scores will closely represent the "true" underlying construct. However, when items unevenly relate to the construct of interest, sum scoring can introduce non-negligible bias into the scores.

Choosing Latent Variable Estimation

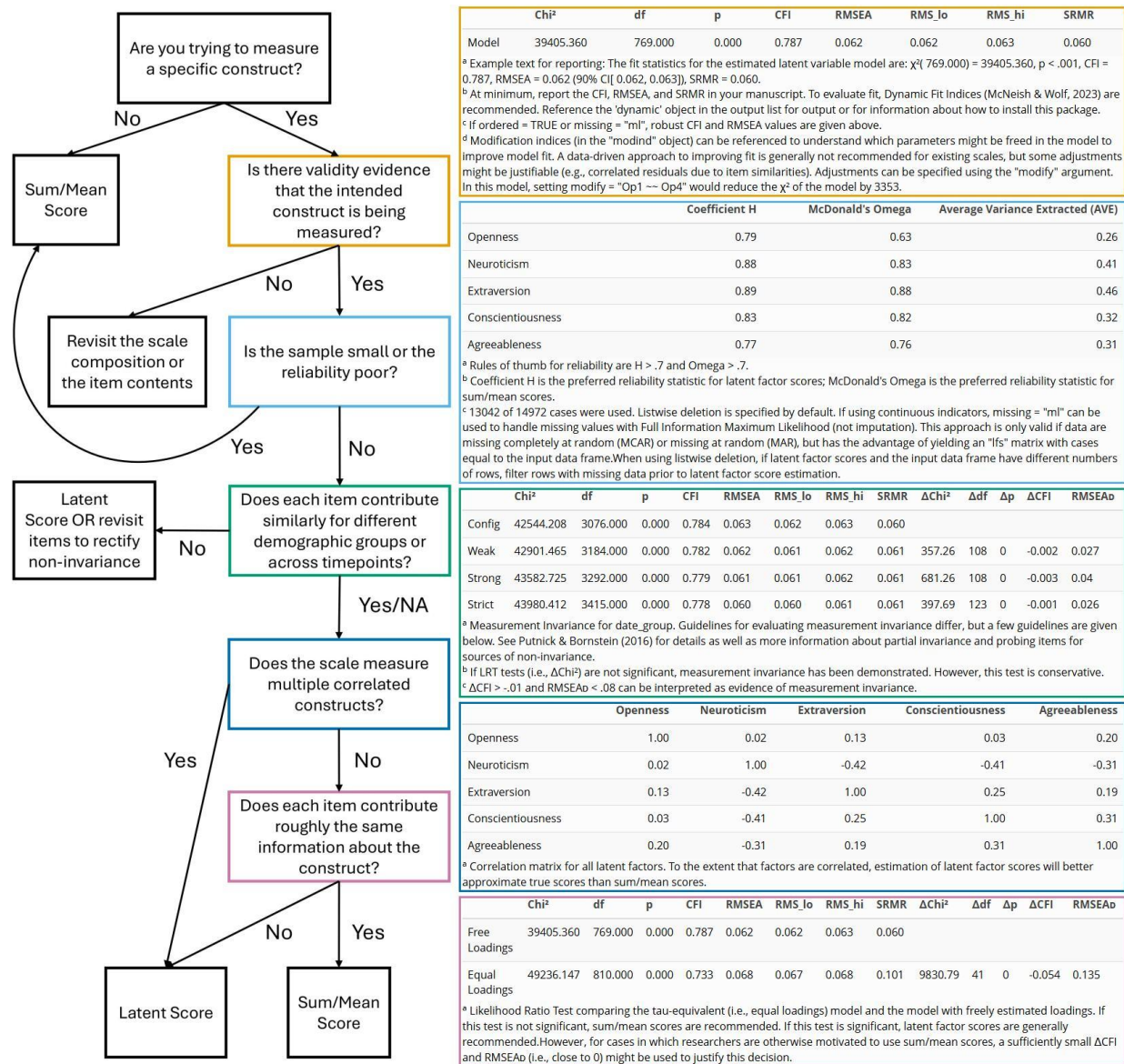
Conversely, latent factor scores tend to be a higher-variance, lower-bias method. The method differentially weights each item in attempt to produce the best possible scores that most closely relate to the "true" construct. However, item-specific weights are based on sample-specific estimates, which have sampling variability. Therefore, the weighting scheme will not be identical for different samples from the same population. In other words, latent factor scoring is more flexible but the penalty for this flexibility is more uncertainty.

In contexts where uncertainty tends to be high such as small sample sizes or low reliability, the uncertainty of latent factor scores can outweigh any potential improvements (Wackwitz & Horn, 1971; Wainer, 1976). That is, latent factor scores may perform better on average, but there is low confidence that they will perform well for a given dataset. In these circumstances, sum scores may be preferable. However, when sample sizes are moderate or large and there is some evidence that items do not necessarily contribute equally to the construct, then the small amount of sampling variability in the weighting scheme may be a worthwhile tradeoff for creating scores with potentially stronger psychometric properties.

The *elfs* Package

We introduce the *elfs* (*Evaluation and Estimation of Latent Factor Scores*) R package, which produces output to help researchers (a) evaluate whether they should use sum scores or latent factor scores and (b) create latent factor scores for use in subsequent statistical models (e.g., regression), if desired. The package removes the need for users to produce their own *lavaan* syntax (Rosseel, 2012) to estimate latent factor scores. To prevent potential issues with misinterpretation, the *elfs* function (within the *elfs* package) provides thorough documentation for interpreting and reporting SEM output. The main vehicle for this documentation is an output image that depicts a flowchart for choosing between sum scores and latent factor scores alongside five color-coded boxes that correspond with the five main decision points in the flowchart. See Figure 1.

Figure 1

Example Output from the *elfs()* Function

Note. The generated image is larger than the one shown above, allowing users to expand the text.

The color scheme follows colorblind-friendly recommendations given in Wong (2011).

At this time, *elfs* documentation is available at <https://sarcyly.github.io/elfs/>. Inputs for the *elfs()* function are:

```
elfs(data, f1_cols,..., fg_cols = NULL, ordered = FALSE, missing =
"listwise", dynamic = FALSE, meas_invar = NULL, modify = NULL,
lfs_method = NULL, lfs_transform = TRUE, chrome_bypass = FALSE,
f1_name = NULL,...)
```

A data frame (“data”) and at least one vector of column names (“f1_cols”) are required. Multi-factor models (up to six factors) can be specified by providing additional vectors of column names (e.g., “f2_cols”). The “fg_cols” argument accepts a vector of column names specifying a general factor for a bifactor model (with a warning that factors are forced to be orthogonal). The “ordered” argument can be set to TRUE to treat indicators as ordinal (continuous indicators use a Maximum Likelihood estimator; ordinal indicators use a Diagonal Weighted Least Squares indicator with robust test statistics). The “missing” argument specifies listwise deletion by default, but other methods can be chosen. The “dynamic” argument omits Dynamic Fit Indices (McNeish & Wolf, 2023) by default, but they are available by request. The “meas_invar” argument can be used to evaluate measurement invariance by including a grouping variable (no restriction on number of groups). The “modify” argument can be used to include terms indicated by modification indices (e.g., a residual covariance due to item similarity). The “lfs_method” argument can be used to specify the latent factor scoring method (e.g., Bartlett, regression). The “lfs_transform” (TRUE by default) transforms the factor scores such that their “mean and variance-covariance matrix matches the model-implied mean and variance-covariance matrix”, as recommended when latent factor scores will be subsequently used in regression analyses (Rosseel, 2012). The “chrome_bypass” argument can be set to TRUE to generate an alternate

output if Google Chrome is unavailable on the given machine. Finally, “fl_name” and subsequent arguments can be used to specify latent factor names (for all defined factors).

The *elfs()* function outputs a list containing:

1. **flowchart**: the primary output as depicted in Figure 2
2. **lfs**: a matrix containing estimated latent factor scores
3. **model**: the *lavaan* model object
4. **summary**: the summary of the *lavaan* model object, with fit statistics, R-squared values, and standardized scores specified
5. **modind**: a sorted list of modification indices for the model
6. **dynamic**: an output object from the *dynamic* package, if requested
7. **fit, rel, tau, corr, meas_invar**: HTML table versions of the tables in **flowchart** from which values can be copy and pasted
8. **model_group, lfs_group**: if measurement invariance testing is requested, lists containing *lavaan* model objects and estimated latent factor scores corresponding with configural, weak, strong, and strict invariance tests

By simplifying the inputs required to generate latent factor models and highlighting validity and reliability indices, we hope to encourage users to report more detailed information about their scales (beyond Cronbach’s α ; see Flake et al., 2017 and McNeish, 2022) and make deliberate choices regarding their scoring method. These outcomes would improve both researchers’ ability to draw conclusions from their models and readers’ ability to evaluate the quality of the measurements reported in scientific articles.

We provide an example of how to use the package, reanalyzing data from Hussey and Hughes (2020) so that structural validity information about the scales is already available.

Example: Need for Cognition Scale

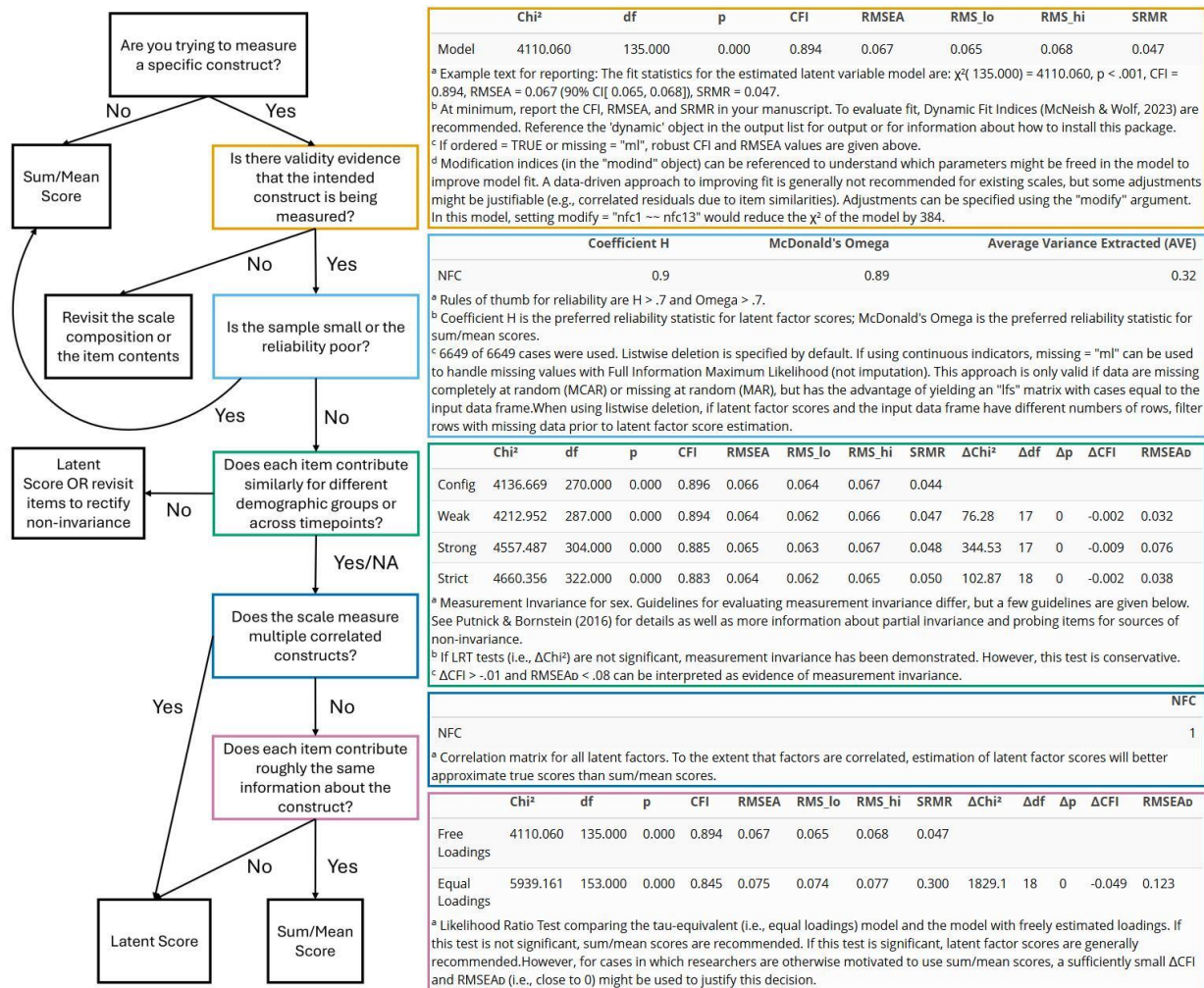
The Need for Cognition Scale (Cacioppo, Petty, & Kao, 1984) includes 18 items intended to capture people's tendency to enjoy and engage in cognitively effortful activities. A markdown of this example is available [here](#). The input code for *elfs* is:

```
nfc_elfs <- elfs(data = nfc_data, f1_cols = starts_with("nfc"),  
dynamic = TRUE, meas_invar = "sex", f1_name = "NFC")
```

The output image is shown in Figure 2. Users can print the **summary** object for a standard *lavaan* summary.

Figure 2

Need for Cognition Scale, elfs Output



Note. The generated image is larger than the one shown above, allowing users to expand the text.

The color scheme follows colorblind-friendly recommendations given in Wong (2011).

Box #1 shows the fit indices for the model: $\chi^2(135) = 4110.06, p < .001, CFI = 0.894, RMSEA = 0.067$ (90% CI[0.065, 0.068]), SRMR = 0.047 (note that this text was copied and pasted from the **fit** object, which is an HTML table). Examination of the **dynamic** object (Figure 3) suggests that a one-dimensional factor model provides a fair approximation of the observed

covariance matrix (95% sensitivity means that models with a “fair” amount of misfit will be rejected 95% of the time using the cutoff in the table), providing one piece of validity evidence supporting that the intended construct is being measured. Were evidence of model fit insufficient, the flowchart would recommend revisiting scale composition or item content because there would be insufficient evidence to claim that the items are reasonably measuring a unidimensional construct.

Figure 3

Dynamic Fit Indices for Need for Cognition

Your DFI cutoffs:					
	MAD	Sim. MAD	CFI	RMSEA	90% CI
Consistent Specificity	0.00	0.00	0.999	0.006	0.008
			95%	95%	
Close Sensitivity	0.038	0.036	0.933	0.052	0.054
			95%	95%	
Fair Sensitivity	0.05	0.047	0.892	0.067	0.069
			95%	95%	
Mediocre Sensitivity	0.06	0.057	0.849	0.082	0.084
			95%	95%	
Estimator: ML					
Sample Size: 6649					

Note. The full **dynamic** output provides additional context regarding sensitivity, evaluation of closeness of fit, mean absolute discrepancy (MAD), and general interpretation.

Box #2 (**rel**) shows that reliability is high, with a coefficient H of .90 and a McDonald's ω of .89, as well as a large sample size of 6649. Coefficient H is appropriate for latent factor scores, whereas McDonald's ω is appropriate for sum scores. Were reliability low or sample size small, the flowchart would recommend sum scores.

Box #3 (**meas_invar**) tests measurement invariance by reported sex. Although we observe significant Likelihood Ratio Tests (examining change in χ^2), this test is conservative and

tests a fairly strict hypothesis of exact invariance, so it is common to supplement Likelihood Ratio Tests with additionally approximate fit measures. Specifically, we instead examine whether $\Delta\text{CFI} > -.01$ and $\text{RMSEA}_d < .08$ for tests of weak, strong, and strict approximate invariance. Both these statistics meet the specified cutoffs for all models, suggesting that approximate measurement invariance by sex holds for the Need for Cognition Scale. Were evidence of measurement non-invariance observed, the flowchart would recommend either latent factor scores (modeling the group differences; see the **lfs_group** object) or probing the items for the source and scale of non-invariance (a reference to Putnick and Bornstein, 2016, is given for this purpose). Note that establishing strict invariance is unnecessary for most purposes.

Box #4 (**corr**) would depict the correlation between latent factors, and modest correlations³ between latent factors would lead to a recommendation of latent factor scores instead of sum scores. However, because Need for Cognition is unidimensional, the flowchart continues to the next box.

Box #5 (**tau**) compares the model fit of the model with free loadings (i.e., the default model) and an essential tau-equivalent model that assumes equal loadings across all items (i.e., the assumed model of sum or mean scores). The essential tau-equivalent model fits significantly worse than the free loadings model, $\Delta\chi^2(18) = 1829.10, p < .001$, suggesting that that items do not contribute equal amounts of information to the Need for Cognition construct. Because these models are nested, differences in fit indices can also be inspected to supplement the $\Delta\chi^2$ test: $\Delta\text{CFI} = -.049$ and $\text{RMSEA}_d = .123$. These values do not meet conventional thresholds to conclude that loadings are approximately equal across all items, but researchers are welcome to interpret the evidence.

³ No specific cutoffs for the size of these correlations have been established at this time.

This last point demonstrates an important theme: following the flowchart and output does not wholly prescribe one scoring type. Instead, the objective is to encourage and facilitate using empirical evidence to select a scoring method, whichever method that may be. Several steps in this process have conventional guidelines for interpretation, but these are merely guidelines and not golden rules or hard cutoffs, so researchers must still weigh the evidence and make decisions that best suit their data and research aims.

Example Writeup for Publication

When researchers use several scales or scales secondary to their main research questions, providing a Markdown or image file with the flowchart might be sufficient. When a scale is of key interest to researchers, we suggest a brief in-text writeup of their scale evaluation and chosen scoring method. Below is an example of how a researcher might summarize the information:

We followed the steps recommended by the *elfs* package ([Hester, Yee, & McNeish, YEAR]) to evaluate the validity of the Need for Cognition Scale and determine an appropriate scoring method. Fit statistics for the scale were $\chi^2(135) = 4110.06, p < .001$, CFI = 0.894, RMSEA = 0.067 (90% CI[0.065, 0.068]), and SRMR = 0.047. Dynamic fit indices (McNeish & Wolf, 2023) indicate that cutoffs of CFI = .892, RMSEA = .067, and upper RMSEA 90% CI = .069 are sensitive to detect misfit consistent with “Fair” model fit; this suggests that model fit is adequate to proceed. Internal reliability is high (coefficient H = .90 [reported if latent factor score chosen]; McDonald’s $\omega = .89$ [reported if sum score chosen]). Analyses support measurement invariance by sex, as increasingly strict models showed values of $\Delta\text{CFI} > -.01$ and $\text{RMSEA}_D < .08$ after invariance constraints were applied (see Putnick & Bornstein, 2016 for a general review of measurement invariance; see Savalei et

al., 2024 for RMSEA_D). A model assuming equal loadings (as sum scores do) fit worse than a model with freed loadings, $\Delta\chi^2(18) = 1829.10$, $p < .001$, $\Delta CFI = -.049$, and RMSEA_D = .123.

[Sum score chosen] However, given our intention to compare scores across multiple studies with different samples, we proceeded with sum scores despite the deterioration in fit.

[Latent factor score chosen] This suggests that differentially weighting items with latent factor scores may be more appropriate and improve psychometric properties, so we proceeded with latent factor scores.

Again, we emphasize that not every choice of scoring method (nor every decision that a psychological scale shows adequate validity and reliability) will be clear-cut; however, a structured report empowers readers and researchers to evaluate the evidence underlying measurement decisions and better communicate these decisions.

Conclusion

The improvement of psychometric practices requires both the creation of new tools and the adoption of existing tools. Psychometricians have created various novel methods to improve measurement models under a wide range of assumptions. However, there has been no “if you build it, they will come” effect, as reviews of psychometric practice consistently show that psychologists do not regularly adopt these methods and instead rely on entrenched heuristic methods that are often taught by graduate supervisors and expected by senior reviewers and editors, despite not always being supported empirically (Aiken, West, & Millsap, 2008; Howard, 2024). This suggests the importance of more direct efforts toward promoting the adoption of new methods. We hope that our primer on selecting an appropriate scoring method, along with the accompanying *elfs* package to ease this process, contributes to a larger movement toward bridging the gap between psychometrics and empirical psychological research.

Open Practices Statement

Documentation of the *elfs* package (and how to download it) can be found at <https://sarcyly.github.io/elfs/> and all code and data for the package can be found at <https://github.com/SarcylyY/elfs>. Submission to CRAN is pending.

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Conflicts of Interest

The authors have no conflicts of interest to report.

Ethics Approval

The present work only involves secondary analysis for the purpose of demonstrating a package's functionality—no ethics approval was required.

Consent to Participate

Not applicable.

Consent for Publication

Not applicable.

Availability of Data and Materials

Documentation of the *elfs* package (and how to download it) can be found at <https://sarcyly.github.io/elfs/> and all code and data for the package can be found at <https://github.com/SarcylyY/elfs>. Submission to CRAN is pending.

Code Availability

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Authors' Contributions

Conceptualization: NH & DM. Methodology: All authors. Software: NH & SY.
Validation: SY. Writing – Original Draft: NH & DM. Writing – Review and Editing: All authors.
Visualization: NH.

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